

Careerbridge AI-Powered Job Matching Platform for Fresh Graduates

Project Vision

The vision of *CareerBridge* is to bridge the gap between education and employment for fresh graduates by providing an intelligent, accessible, and affordable job-matching platform. The system focuses on empowering new graduates to discover career opportunities that align with their skills, qualifications, and interests while enabling employers to identify suitable entry-level candidates efficiently. Through AI-based matching algorithms and data-driven recommendations, *CareerBridge* aspires to become a one-stop platform that enhances employability, streamlines recruitment, and supports fresh graduates to start their career.

Scope and Features:

- AI-powered job and candidate matching using skill-based algorithms.
- Profile creation with resume upload, skills, and career interests.
- Smart job recommendations based on similarity scoring (rule-based + TF-IDF).
- Employer dashboard for job posting, candidate shortlisting, and analytics.
- Admin dashboard for user management and system monitoring.
- Email notifications and status tracking for applications.
- Responsive design for both desktop and mobile users.

Problem Statement

Despite the rising number of graduates in India, youth unemployment remains a pressing challenge due to poor visibility of opportunities and lack of efficient job skill matching systems. Existing job portals such as LinkedIn and Naukri are optimized for experienced professionals, making it difficult for fresh graduates to stand out. Manual searches through unfiltered listings are time-consuming and inefficient. Small companies also struggle to reach suitable entry-level candidates due to high platform costs.

Hence, there is a strong need for a dedicated AI-powered job-matching platform that focuses exclusively on connecting graduates with relevant job opportunities and helps employers easily identify new talent.

Proposed Solution

CareerBridge offers an AI-driven solution that automates the process of connecting job seekers and recruiters through intelligent skill mapping and similarity-based ranking algorithms.

The platform allows:

- **Graduates** to register, upload resumes, specify skills and preferences, and receive job recommendations.
- **Employers** to post job details and automatically receive candidate matches ranked by relevance.

- **Admins** to manage users, employers, and system analytics efficiently.

Developer Features:

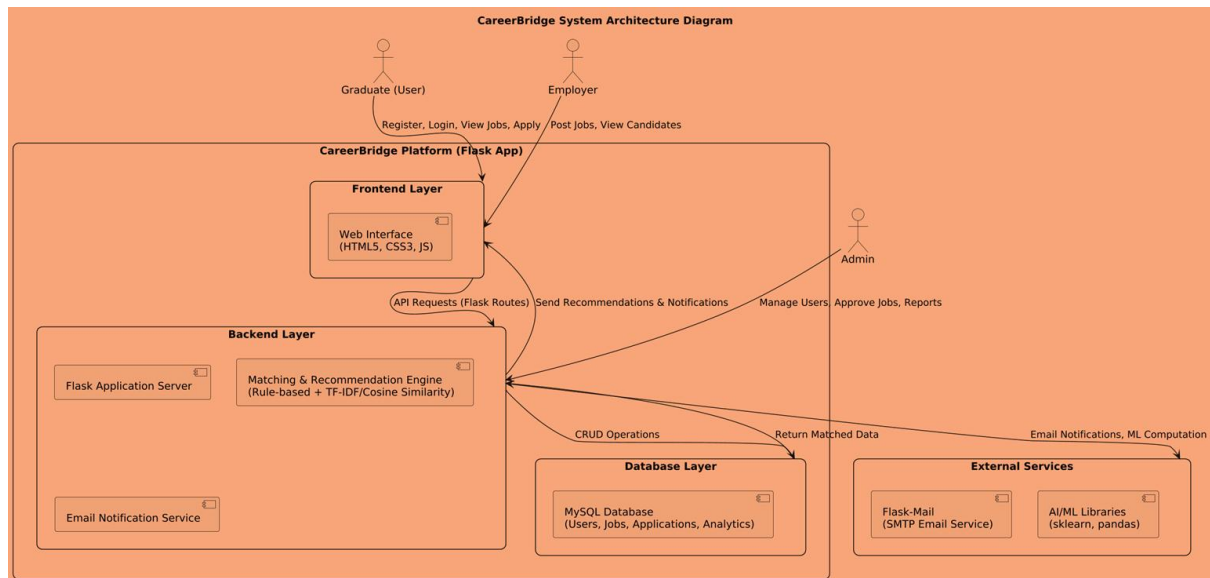
- **Skill Matching Algorithm:** Uses rule-based weighted similarity to compute match scores between candidate and job skills.
- **TF-IDF & Cosine Similarity:** For textual comparison between resumes and job descriptions.
- **Automated Ranking:** Considers skill match, recency, and company credibility.
- **Analytics Dashboard:** Displays employment statistics, job trends, and user activity.
- **Scalability:** Designed for expansion into mobile apps and integration with AI-based interview tools.

This intelligent automation ensures faster, fairer, and more personalized job matching — effectively bridging the gap between graduates and recruiters.

Risk Plan

Risk	Description	Impact	Mitigation Strategy
Algorithm Accuracy	Matching algorithm may not provide accurate or relevant job recommendations.	High	Conduct multiple testing iterations; fine-tune similarity thresholds; gather feedback from users for algorithm improvement.
System Downtime or Server Failure	Hosting issues or unexpected server crashes could interrupt access to the platform.	High	Implement cloud-based backup servers and periodic database backups; use reliable hosting services.
Integration Failures	Problems integrating the Flask backend with the MySQL database or external APIs.	Medium	Conduct modular testing and continuous integration; maintain version control using Git.
Scalability Issues	Increasing users may lead to slower response times or database overload.	Medium	Optimize queries and database indexing; deploy load balancers or scalable cloud infrastructure.

High-Level Diagram



Literature Review

1. Çelik Ertuğrul, D. (2025) – Job recommender systems: a systematic literature review (SLR)

Source: Journal of Big Data

Link: <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-025-01087-9>

Summary: This paper surveys job recommender systems from the last decade, classifying methods (content-based, collaborative, hybrid, skill-aware) and listing common evaluation metrics and datasets. It highlights that hybrid models often perform best.

Relevance to CareerBridge: Useful for choosing a hybrid architecture and evaluation metrics to measure CareerBridge's performance.

2. Rosenberger J. et al. (2025) – CareerBERT: Matching resumes to ESCO jobs in a shared framework

Source: Elsevier / ScienceDirect

Link: <https://www.sciencedirect.com/science/article/pii/S2666827025000081>

Summary: Proposes using transformer-based language models (BERT variants) fine-tuned on resume/job corpora to improve semantic matching between resumes and job descriptions.

Relevance to CareerBridge: Suggests upgrading TF-IDF resume matching to transformer embeddings for better semantic understanding.

3. JobFormer: Skill-Aware Job Recommendation with Transformer Models (ACM, 2024)

Source: ACM Digital Library

Link: <https://dl.acm.org/doi/10.1145/3637459.3641031>

Summary: Presents a transformer-based architecture that models skills and career trajectory to recommend jobs. Improves ranking of suitable entry-level jobs.

Relevance to CareerBridge: Blueprint for building a skill-aware matching layer in CareerBridge for better recommendations.

4. Sarsenbay, S. et al. (2025) – Generating Job Recommendations Based on User Traits and Behavior

Source: MDPI Algorithms

Link: <https://www.mdpi.com/1999-4893/18/3/110>

Summary: Extends job recommenders by adding user traits and behavioral signals to improve personalization.

Relevance to CareerBridge: Suggests including user behavior data in CareerBridge's ranking for personalized results.

5. Resume Analyzer and Job Recommendation System (IRE Journals / IRJMETS / IJRPR / IJIRT, 2024–2025)

Source: Various Applied Engineering Journals

Link: <https://irejournals.com/formatedpaper/1705155.pdf>

Summary: Describes TF-IDF and cosine similarity-based resume parsers for fast, explainable entry-level job matching.

Relevance to CareerBridge: Confirms TF-IDF + cosine is a valid, low-cost baseline for CareerBridge's MVP.

6. Skill-Based Job Matching – A Novel Approach (ScienceDirect, 2025)

Source: ScienceDirect

Link: <https://www.sciencedirect.com/science/article/pii/S1877050924004472>

Summary: Combines skill extraction and weighting rules to produce better match scores and reduce false positives.

Relevance to CareerBridge: Supports CareerBridge's use of a weighted skill-based matching algorithm.

7. Graduate Employability and Skill Gap in India (2020–2025 Reports)

Source: NIEPA / IJFMR / OASIS Publications

Link: https://www.researchgate.net/publication/368521688_Graduate_Employability_in_India

Summary: Discusses the persistent gap between graduate education and employment in India and the need for dedicated job platforms.

Relevance to CareerBridge: Validates CareerBridge's problem statement and focus on fresh graduates.

8. Hybrid Job Matching Frameworks and Evaluation Metrics (2024–2025)

Source: Multiple Research Studies

Link: <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-025-01087-9>

Summary: Recommends combining fast TF-IDF baseline models with transformer re-ranking for improved performance.

Relevance to CareerBridge: Provides a roadmap for CareerBridge's future upgrade strategy.

Literature Review (Clarity and Project Initiation)

Several studies and systems have explored AI and data-driven methods to improve recruitment and job recommendation processes. However, most existing approaches focus on experienced professionals rather than fresh graduates.

1. Existing Research and Systems

- *Kaur & Sharma (2021)* proposed AI-driven recruitment systems that use machine learning algorithms to screen and shortlist candidates automatically. Their work demonstrates how automation can reduce manual recruitment time.
- *Patil et al. (2022)* developed a job recommendation framework using TF-IDF and cosine similarity to compare resumes with job descriptions, showing improved accuracy in candidate-job matching.
- *Zhang et al. (2023)* introduced hybrid recommendation systems combining rule-based and similarity-based algorithms to improve match relevance.
- *Banerjee & Das (2022)* discussed the need for localized employment platforms that support small businesses and fresh graduates, particularly in developing countries.

2. Gap Identified

Most existing systems:

- Target experienced professionals, not fresh graduates.
- Lack personalized skill-based recommendations.
- Often require paid access for better job visibility.
- Do not provide analytics for employers or trend insights.

3. How *CareerBridge* Differs

The CareerBridge platform addresses these gaps by:

- Focusing exclusively on fresh graduates and entry-level positions.
- Implementing a rule-based skill-matching algorithm combined with TF-IDF-based text similarity to recommend jobs accurately.
- Offering free and accessible tools for both graduates and small companies.
- Including admin dashboards and analytics for monitoring job trends and improving decision-making.

Gantt chart:

